

MAINCRAFTS TECHNOLOGY | TASK 01

California Housing Price Prediction

AI & Machine Learning Internship Portfolio

01. OVERVIEW

Project Objectives & Technical Scope

| Internship Task Objectives



Technical Workflow

Implementing the end-to-end ML lifecycle: from data fetching and EDA to model training and evaluation using Scikit-Learn.



Predictive Goal

Building a robust **Linear Regression** model to predict median house values based on census-level demographic features.

02. DATA AUDIT

Exploratory Data Analysis (EDA)

| Dataset Characteristics

Structural Profile

-  **20,640 Samples:** Comprehensive census data from 1990 California survey.
-  **Clean Data:** Zero missing or null values identified across all 8 features.
-  **Key Features:** Median Income, House Age, Avg Rooms, and Geographic Coordinates.

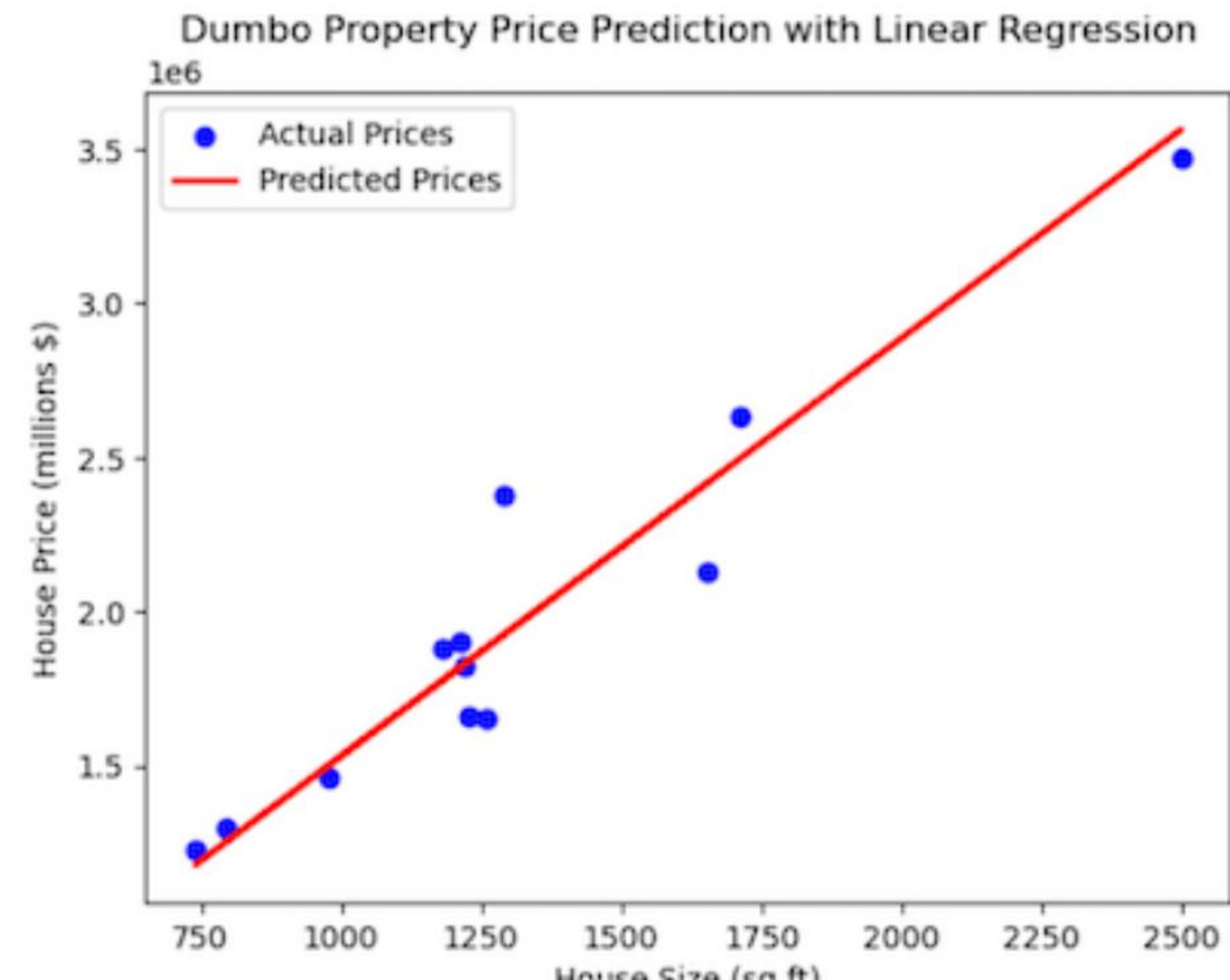
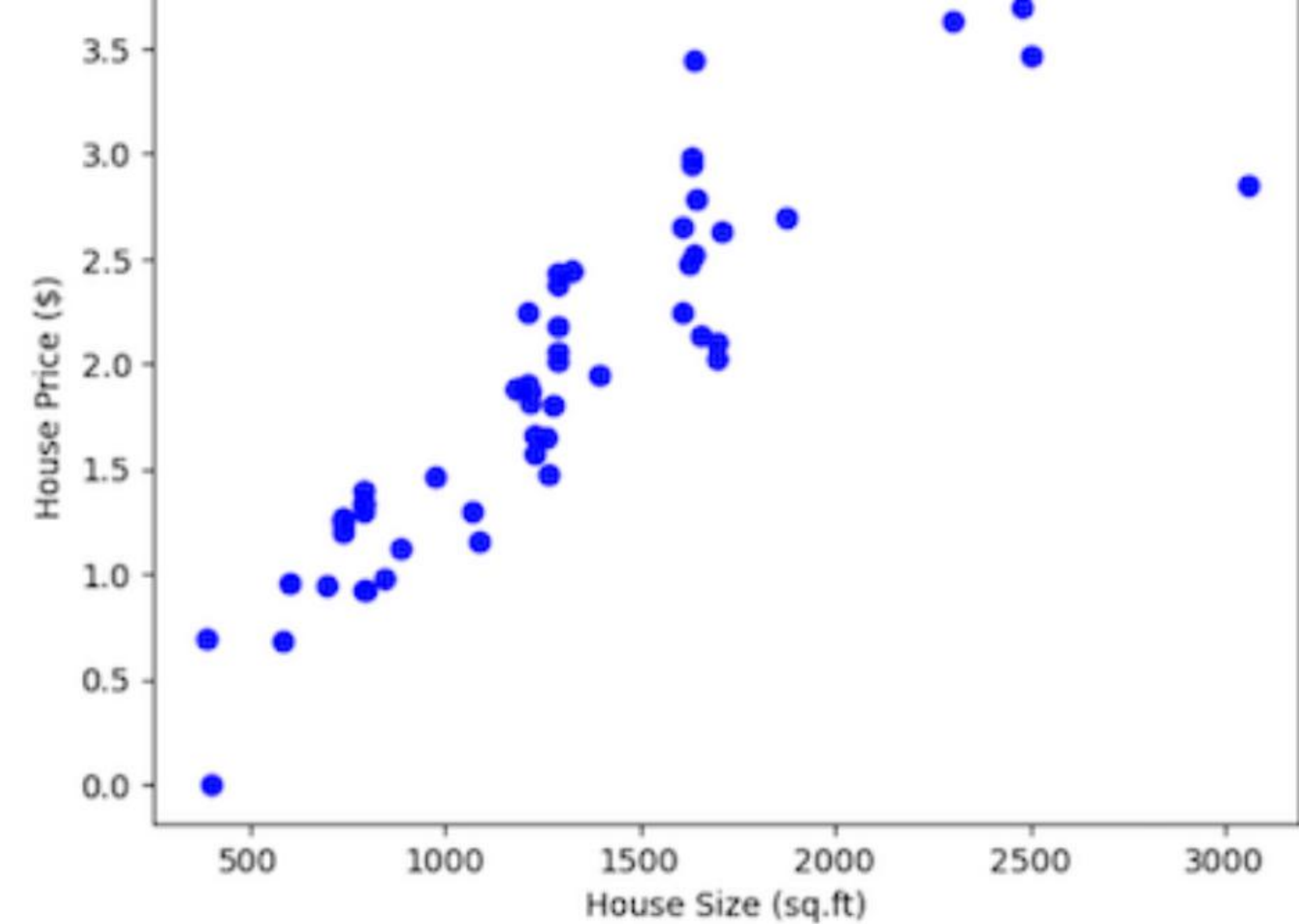


| Correlation Analysis

The Income-Price Link

During analysis, **Median Income (MedInc)** emerged as the strongest predictor with a correlation of **0.69**.

Interestingly, the target variable is artificially capped at **5.0 (\$500k)**, creating a unique challenge for linear modeling as luxury data is truncated.



03. EVALUATION

Model Training & Benchmark Results

| Primary Performance Metrics

0.53

MAE (\$100k)

0.75

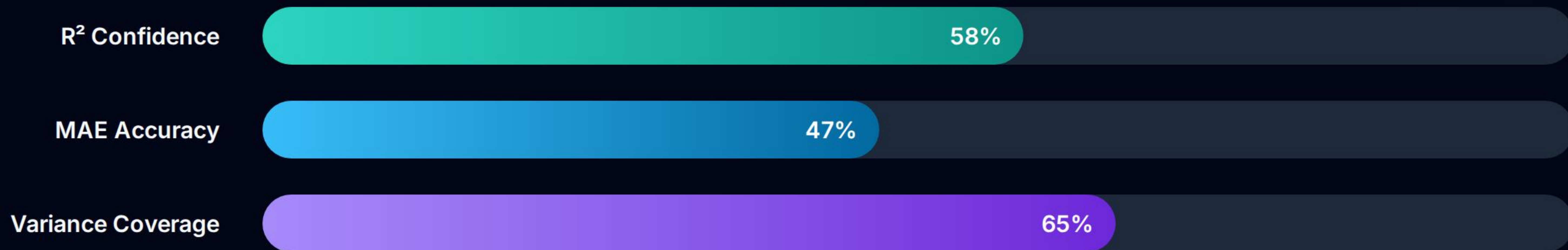
RMSE

58%

R² SCORE

The baseline model explains approximately 57.6% of the price variance across districts.

Metric Distribution Analysis



Comparison of model statistical confidence versus actual error margins in standard units.

Residual Distribution Trend



The model exhibits heteroscedasticity due to the target variable cap at 5.0.

| Strategic Roadmap

- ✂ **Feature Engineering:** Calculating spatial proximity to high-value coastal hubs like San Francisco and LA.
- 📦 **Advanced Ensembles:** Transitioning to non-linear models such as **XGBoost** and **Random Forest**.
- 🔍 **Outlier Management:** Addressing truncated data caps to improve prediction accuracy for high-end real estate.
- 🔗 **Model Deployment:** Saving weights via Pickle/Joblib for integration into a real-time web UI.



Thank You!

MAINCRAFTS TECHNOLOGY - TASK 01 COMPLETE

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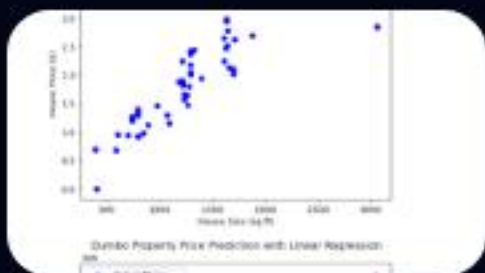
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<https://raw.githubusercontent.com/codedex-io/projects/main/projects/predict-home-prices-with-python-and-linear-regression/graph.png>

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